

# Retail Store Intelligence Challenge

HACKATHON SERIES

JUNIOR ENGINEERS TRACK

A structured, progressive hackathon designed for junior software engineers at a large retail chain. Participants will automate real-world retail operations — from billing and discounts to full-day performance dashboards — using core Java programming concepts. Each level mirrors an actual business scenario, bridging the gap between code and commerce.

**Developed by Talenciglobal** — Empowering the next generation of enterprise software engineers through industry-aligned, challenge-based learning.

# Why Retail Automation Matters

The global retail technology market is valued at over **\$238 billion** as of 2024, with automation and intelligent billing systems accounting for a rapidly growing share of that investment. Retailers who deploy automated billing and analytics tools report up to **30% reduction in checkout errors** and a **25% increase in operational efficiency**. Understanding how to translate business logic into code is no longer optional for engineers in this sector — it is foundational.

This hackathon is structured around five progressive levels, each simulating a real department within a retail organization. Participants move from a single-product billing assistant to a full-branch performance dashboard, mirroring the actual complexity junior engineers encounter in enterprise retail environments.

5

Challenge Levels

From beginner billing to expert KPI dashboards

\$238B

Retail Tech Market

Global market size driving demand for automation engineers

30%

Error Reduction

Average drop in billing errors with automated systems

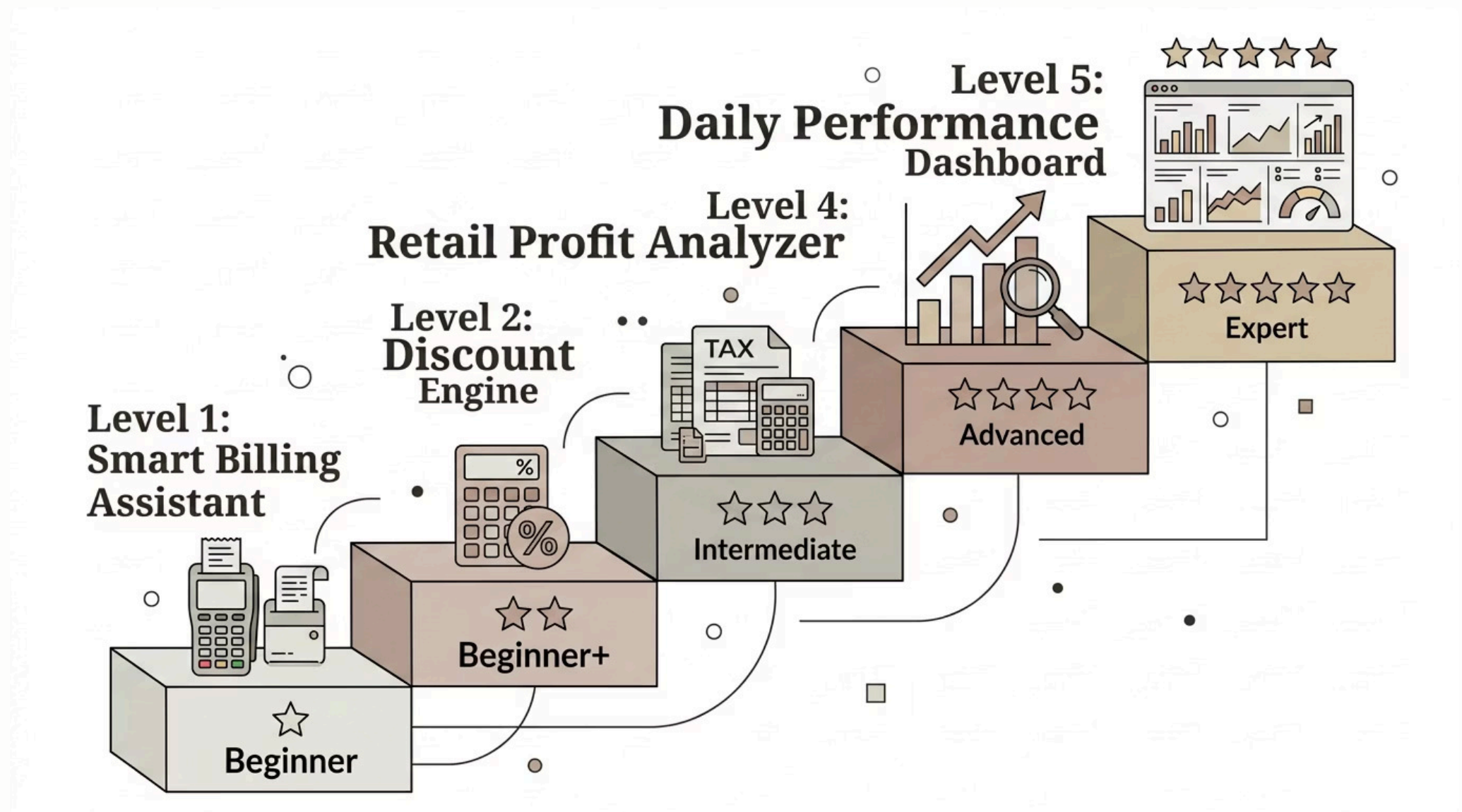
25%

Efficiency Gain

Operational improvement reported by retail automation adopters

# Challenge Roadmap: Five Levels of Mastery

The challenge is structured as a progressive difficulty ladder. Each level builds upon the concepts introduced in the previous one, ensuring a coherent learning journey. Participants are expected to complete levels sequentially, as each level introduces variables and patterns that reappear in more complex forms at higher levels.



According to industry training benchmarks, engineers who complete challenge-based learning modules that mirror real business contexts demonstrate **40% higher code readability scores** and are **2x more likely** to pass technical screening rounds compared to peers trained through abstract exercises alone.

# Level 1: Smart Billing Assistant

★ BEGINNER

Every retail operation begins at the billing counter. In this foundational challenge, participants simulate the most basic function of any point-of-sale system: computing the total cost of a customer's purchase. A customer purchases a single product, and the system must calculate the final payable amount based on the unit price and quantity purchased.

## Inputs Required

- **Product Name** — String variable to store item identity
- **Unit Price** — Integer representing cost per unit
- **Quantity** — Integer representing number of units purchased

## Expected Output

- Product Name printed to console
- Total Cost = Unit Price × Quantity

## Example Run

**Input:** Rice, 50, 4

**Output:** Rice 200

## Core Concepts Tested

- String variable declaration and usage
- Integer variable assignment
- Multiplication operator in Java
- Formatted output using `System.out.println()`

## Thinking Skill

Understanding how real-world information — a product label, a shelf price, a purchase quantity — maps directly to named variables in code.

# Level 2: Discount Engine

★★ BEGINNER+

Retail discounting is one of the most powerful levers in consumer psychology. Studies show that **93% of shoppers** use a coupon or discount offer at least once per year, and promotional pricing events can drive a **15–45% spike in transaction volume**. Behind every displayed discount is a precise calculation — and that calculation must be coded correctly to protect margin.

In this level, participants build a discount engine that accepts a product's original price and a discount percentage, then computes the discount amount and the final payable price. This introduces float variables and the importance of mathematical expression accuracy in financial systems.

## Inputs Required

- **Product Price** — Float or integer representing original price
- **Discount Percentage** — Float representing the offered discount rate

## Expected Output

- $\text{Discount Amount} = \text{Price} \times (\text{Discount\%} / 100)$
- $\text{Final Price} = \text{Price} - \text{Discount Amount}$

## Example Run

**Input:** 2000, 10

**Output:** 200 | 1800

## Core Concepts Tested

- Float variable declaration
- Percentage-based arithmetic expressions
- Intermediate variable storage
- Multiple outputs in a single program

## Thinking Skill

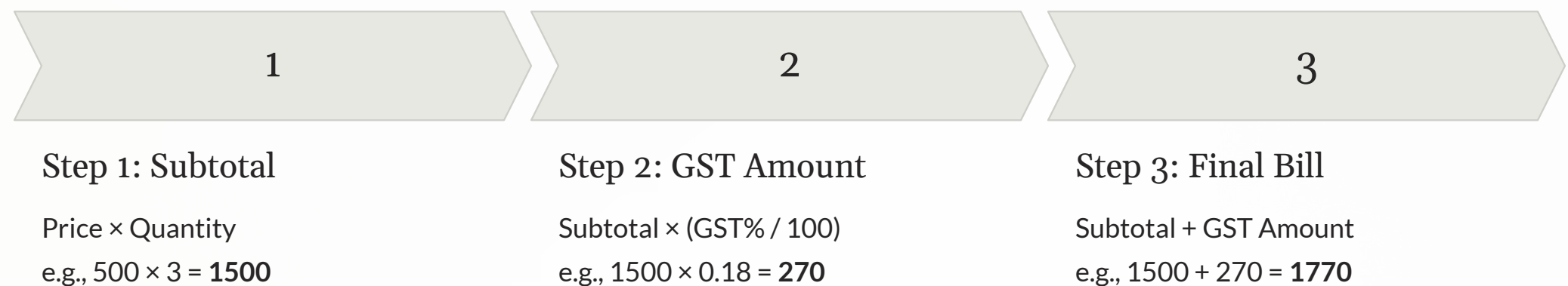
Converting a business rule — "10% off the listed price" — into a precise, unambiguous mathematical expression that a computer can evaluate reliably.

# Level 3: GST Billing System

★★★ INTERMEDIATE

Taxation is a non-negotiable component of any retail billing system. In India alone, over **1.4 crore GST-registered businesses** are required to compute and display tax amounts on every invoice. Globally, value-added tax and goods-and-services tax frameworks make accurate tax calculation a legal obligation — not merely a feature. Errors in tax computation can result in compliance penalties and audit risk.

At this level, participants build a GST-compliant billing module. The program accepts a product price, quantity, and applicable GST percentage, then computes the subtotal, the GST amount, and the final invoice total. This challenge introduces the concept of **expression chaining** — where one computed result feeds directly into the next calculation.



## Core Concepts Tested

- Multiple variable declarations and assignments
- Intermediate result storage in named variables
- Chained arithmetic expressions
- Real-world financial calculation modeling

## Thinking Skill

The critical insight at this level is **decomposition** — breaking a larger financial calculation into smaller, labeled intermediate steps rather than attempting to resolve it in a single complex expression. This mirrors professional coding standards in enterprise billing software.

# Level 4: Retail Profit Analyzer

★★★★ ADVANCED

Profitability analysis is the backbone of retail management decision-making. According to McKinsey, **retail stores that implement systematic margin tracking at the SKU level** outperform competitors by up to **8 percentage points in annual profit growth**. Store managers need clear, accurate data on which products are generating returns and which are eroding margin. This level simulates exactly that analytical function.

Participants receive a product's cost price, selling price, and units sold. They must compute revenue, investment, profit, and profit percentage — outputting each as a labeled result. The challenge is deliberately structured to expose a common failure pattern: students who attempt to calculate everything in a single giant expression consistently produce incorrect outputs. This level **forces procedural decomposition** as a mandatory problem-solving strategy.

## Revenue

Selling Price  $\times$  Units Sold

**Example:**  $1000 \times 50 = 50,000$

## Investment

Cost Price  $\times$  Units Sold

**Example:**  $800 \times 50 = 40,000$

## Profit

Revenue  $-$  Investment

**Example:**  $50,000 - 40,000 = 10,000$

## Profit Percentage

$(\text{Profit} / \text{Investment}) \times 100$

**Example:**  $(10,000 / 40,000) \times 100 = 25\%$



**Common Failure Pattern:** Students who calculate everything in a single nested expression frequently produce wrong results due to operator precedence errors. This level is specifically designed to teach the discipline of storing intermediate results in named variables before using them in subsequent calculations.

# Level 5: Daily Performance Dashboard

★★★★★ EXPERT

This is the capstone challenge. A retail branch manager requires a single-screen summary of the day's business performance — a dashboard composed of five interconnected Key Performance Indicators. According to Deloitte's Retail Operations Survey, **branch managers who review daily KPI dashboards** make inventory and staffing decisions that are on average **35% more accurate** than those relying on end-of-week summaries. Real-time and daily performance visibility is now considered a baseline operational requirement in modern retail.

What makes this level uniquely challenging — and uniquely educational — is the **data dependency chain**. Each computed output becomes the input for the next calculation. This is the first moment in the challenge series where participants directly experience a foundational principle of enterprise software design: sequential, dependent computation pipelines.



This dependency structure is the defining intellectual challenge of Level 5. Students who attempt to compute all five KPIs simultaneously — without understanding that Output A becomes Input B — will produce systematically incorrect results. Mastering this level signals readiness for real-world enterprise development tasks.

# Level 5: Formula Reference & Example Run

The following table provides the complete formula set for the Daily Performance Dashboard, along with the example input and expected output values. Participants should verify each intermediate result before proceeding to the next calculation. This disciplined approach to sequential validation is standard practice in financial software development.

| KPI              | Formula                              | Example Input            | Expected Output |
|------------------|--------------------------------------|--------------------------|-----------------|
| Gross Sales      | Customers × Avg Purchase Value       | 500 × 1200               | 600,000         |
| Discount Amount  | Gross Sales × (Discount% / 100)      | 600,000 × 10%            | 60,000          |
| Net Sales        | Gross Sales – Discount Amount        | 600,000 – 60,000         | 540,000         |
| Operating Profit | Net Sales – Operating Cost           | 540,000 – 450,000        | 90,000          |
| Profit Margin %  | (Operating Profit / Net Sales) × 100 | (90,000 / 540,000) × 100 | 16.67%          |

 Industry Benchmark: A profit margin of 16.67% on net sales is considered healthy for a mid-sized retail branch. The national average net profit margin for general merchandise retail in India ranges between 3% and 18%, depending on product category and operational efficiency.

|                           |                                      |                                    |                         |
|---------------------------|--------------------------------------|------------------------------------|-------------------------|
| Customers<br>500 walk-ins | Avg Purchase<br>₹ 1,200 per customer | Operating Cost<br>₹ 4,50,000 total | Discount<br>10% applied |
| Margin<br>16.67% achieved |                                      |                                    |                         |

# Evaluation Criteria & Closing Notes

DEVELOPED BY TALENCIAGLOBAL

Participants will be evaluated across five dimensions that mirror the standards applied in enterprise Java development environments. The hackathon is not merely a test of whether the correct numeric output is produced — it is an assessment of professional coding discipline, logical clarity, and the ability to translate business requirements into maintainable, well-structured code.



## Correctness

All outputs must match expected values exactly. Partial credit is awarded for correct intermediate calculations even if the final output contains errors.



## Decomposition

Calculations must be broken into labeled intermediate steps. Single-expression solutions that happen to produce correct output will not receive full marks.



## Code Clarity

Variables must be meaningfully named. Programs must be readable by a peer engineer without requiring explanation from the author.



## Business Alignment

Output labels must match the business terminology specified. Variable names should reflect the domain context, not generic placeholders like `x` or `temp`.

The greatest engineers are not those who write the most clever code — they are those who write code that a business stakeholder and a fellow engineer can both understand. This hackathon is designed to develop exactly that discipline. —

Talenciaglobal Engineering Education Team